

## Research Profile

*I am a postdoctoral researcher in psychiatry and translational neuroscience building an independent program in biomimetic neuromodulation. My research focuses on designing brain-derived stimulation waveforms that preserve endogenous temporal structure and validating target engagement with electrophysiology (i.e., high density EEG) to guide rational optimization of novel stimulation treatments. I have recently developed transcranial endogenous current stimulation (tECS), physiology-informed TMS (piTMS), and my recent work has involved leading clinical trials using closed-loop tACS in major depressive disorder and examining intracranial electrophysiology in mood circuits.*

## Education and Training

### Doctor of Philosophy (PhD), Clinical Neuroscience

Oct 2020 – Apr 2024 (Conferred)

The University of Queensland Child Health Research Centre, South Brisbane.

Thesis: *"Investigating the use of non-invasive brain stimulation to improve attention in paediatric acquired brain injury"*

Advisor: Prof. Karen Barlow PhD, MSc, MBChB, MRCPCH, FRACP

- Conducted mechanistic investigations HD-EEG, fMRI, and DTI/NODDI, as well as two clinical trials to study mechanisms of recovery and response to non-invasive brain stimulation in pediatric brain injury.
- Examined structural and functional biomarkers associated with interindividual variability in tDCS response.
- Developed and implemented a home-based, remotely supervised tDCS clinical trial in pediatric brain injury.

### Master of Public Health (Global Health)

Jan 2019 – Jul 2020

The University of Queensland, Herston

Thesis: *"Access to Deep Brain Stimulation Treatment for Patients with Parkinson's Disease: A Comparative Policy Analysis of Fourteen High-Income Countries"* (see publications)

- HRH Duchess of Gloucester Prize in Global Health (2020)

### Bachelor of Biomedical Science, Neuroscience

Jan 2016 – Dec 2018

The University of Queensland, St Lucia;

- Queensland Brain Institute Research Intern, high distinction (2018)
- School of Chemistry Undergraduate Research Program, high distinction (2016)

## Academic Appointments

### Postdoctoral

Jul 2024 – present

University of North Carolina at Chapel Hill (Reference: Flavio Frohlich, [flavio\\_frohlich@med.unc.edu](mailto:flavio_frohlich@med.unc.edu))

- Developed biomimetic neuromodulation approaches in psychiatry, including transcranial endogenous current stimulation (tECS), physiology-informed TMS (piTMS)
- Leading scientific development and implementation of clinical trials across three translational neuromodulation studies, including closed-loop tACS in major depressive disorder and intracranial electrophysiology projects.
- Conducting high-density EEG and source-localized connectivity analyses to define target engagement mechanisms.
- Primary mentor/advisor for 9 undergraduate research assistants and 3 full-time study coordinators (clinical trial team)

### Senior Research Assistant

Jul. 2020 – Jul 2024

University of Queensland Child Health Research Centre (Reference: Karen Barlow, [k.barlow@uq.edu.au](mailto:k.barlow@uq.edu.au))

- rTMS and EEG technician in pediatric neuromodulation and neurophysiology research across concussion, autism, and acquired brain injury studies

### Senior Research Assistant

Aug 2022 – Dec 2022

UQ School of Public Health (Reference: Marina Reeves, [m.reeves@sph.uq.edu.au](mailto:m.reeves@sph.uq.edu.au))

- Mixed methods analysis of longitudinal data on student feedback following redesign of PUBH7000 (Research Methods in MPH degree)

**Contracted Academic Editor****Jul. 2019 – Oct 2020****Enago** (Reference: Veronica Jacob, veronicaj@enago.com)**Uni-edit** (Reference: Daniel Boles, daniel.uniedit@gmail.com)

- **Specialisations:** biomedical science; biology; neuroscience; public health; social science; humanities; policy
- Edited over 70 manuscripts for publishing in international journals including BMJ open; PLOS ONE; AJRCMB

**Summer Research Scholarship recipient****Nov. 2019 – Feb. 2020****UQ Child Health Research Centre** (Reference: Kartik Iyer, k.iyer@uq.edu.au)

- Led a task fMRI analysis of working memory in 90 children with acquired brain injury and healthy controls

**Teaching Experience****Teaching Assistant, PUBH7000 (Research Methods for Master of Public Health)****Feb. – Jun. 2021- 2023**

The University of Queensland (Reference: Marina Reeves, m.reeves@sph.uq.edu.au)

- Facilitating 3 hour weekly workshops
- Delivered guest lectures and workshops each semester
- *Received a score of 5/5 for SECAT student teaching evaluations each year 2021- 2023*
- *Appointed **Fellow of the Higher Education Academy (FHEA), 2021: Demonstrate substantial, evidence-based teaching experience in higher education, aligned with the [Professional Standards Framework](#)***

**Co-Course Coordinator, BIOM1061 (Pharmacology for Nursing and Midwifery)****Nov. 2020 – Jan 2021**

The University of Queensland (Reference: Mary-Louise Manchadi, m.roymanchadi@uq.edu.au)

- Co-coordinated summer course

**Tutor and course development consultant, BIOM3200 (Biomed. Sci. Capstone Course)** **Aug. – Oct. 2020; 2021**

The University of Queensland (Reference: Jo Bowles, jo.bowles@uq.edu.au)

- Played a lead role in re-development of Bachelor of Biomedical Science Research Capstone course

**Research Funding (USD ~\$435,000)**

| Date                                | Funding Source                                                                    | Role / Purpose                                                  | USD       | Original (AUD) |
|-------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------|----------------|
| <b>Research Grants</b>              |                                                                                   |                                                                 |           |                |
| Apr 2026                            | Foundation of Hope: Impact Grant                                                  | Co-PI, piTMS clinical trial                                     | \$200,000 |                |
| Jan 2025                            | Brain Behavior Research Foundation (BBRF) NARSAD Distinguished Investigator Grant | Co-I; writing lead; tECS clinical trial                         | \$100,000 |                |
| Nov 2021                            | Small Equipment Grant, UQ Child Health Research Centre                            | Primary applicant; funded fMRI-compatible tDCS/tACS device      | \$6,600   | \$10,000       |
| Sep 2021                            | Small Research Grant, UQ Child Health Research Centre                             | Primary applicant; study costs                                  | \$3,300   | \$5,000        |
| Aug 2021                            | Pilot Scheme, Herston Imaging Research Facility (HIRF)                            | Co-I; awarded 10 MRI scans                                      | \$4,290   | \$6,500        |
| <b>Fellowships and Scholarships</b> |                                                                                   |                                                                 |           |                |
| Jul 2024                            | American Australian Association Graduate Education Scholarship                    | Primary sole applicant                                          | \$40,000  |                |
| Oct 2020 - Apr 2024                 | Australian Government Research Training Program (RTP) Scholarship                 | Living stipend (approx. \$32,000 AUD per year for 3.5 years)    | \$73,920  | \$112,000      |
| <b>Travel Awards</b>                |                                                                                   |                                                                 |           |                |
| Jan 2026                            | Society for Biological Psychiatry                                                 | Travel Award to SOBP 2026 NYC                                   | \$2,000   |                |
| Sep 2023                            | UQ Faculty of Medicine Thomas McIntyre Scholarship                                | Travel to ABSS and ACNS (Sydney, Australia)                     | \$1,310   | \$1,990        |
| Feb 2023                            | UQ Faculty of Medicine Researcher Development Award                               | Travel to International Brain Stimulation Conference (Portugal) | \$2,640   | \$4,000        |
| Aug 2022                            | Abbott Science and Society Award for Emerging Scientists                          | Travel to NYC Neuromodulation                                   | \$1,000   |                |

## Honors and Awards

### Academic Awards and Fellowships

- Society for Biological Psychiatry (SOBP) Travel Award Recipient (2026)
- Runner Up, PhD Graduate of the Year, UQ Faculty of Medicine (2024)
- Student of the Year, UQ Child Health Research Centre (2023)
- HRH Princess Alice, Duchess of Gloucester Prize in Global Health – Highest grade in Global Health (2020)
- Dean's Academic Commendation, UQ School of Public Health (2020)
- UQ Summer Research Scholarship – Child Health Research Centre (2019–2020)

### Public Engagement and Teaching Recognition

- Fellow of the Higher Education Academy (FHEA) – UK Professional Standards Framework (2021)
- Teaching and Learning Excellence Award, UQ Faculty of Medicine (2021)

### Student Competitions and Popular Vote Awards

- People's Choice Presentation, UniQuest Commercialisation Workshop (2023)
- People's Choice Presentation, UQ Three Minute Thesis School/Institute Final (2023)
- Highly Commended Presentation, UQ CHRC HDR/EMCR Showcase (2023)
- Best Runner Up, UQ Faculty of Medicine HDR Symposium (2021)

## Selected Invited Talks and Presentations

### Invited Talks and Seminars

1. Invited Speaker — American Australian Association Annual Benefit Dinner, New York City, USA (Oct 2024)
2. Invited Speaker — UC & Western Neurotrauma Symposium, UCLA, Los Angeles, USA (Oct 2024). "Developing tDCS as a treatment for attention problems in pediatric acquired brain injury: From biomarker identification to at-home treatment"
3. Invited Seminar — UC San Diego (USA) and CAMH Toronto (Canada) (Jul 2022). "Using non-invasive brain stimulation to improve executive function following acquired brain injury in children"

### Symposia and Oral Presentations

4. Symposium Chair, Organizer, and Speaker — 6th International Brain Stimulation Conference, Kobe, Japan (Feb 2025). "At-home brain stimulation in complex clinical populations: Practical, clinical, and scientific considerations"
5. Symposium Co-Chair and Organizer — 6th International Brain Stimulation Conference, Kobe, Japan (Feb 2025). "Closed-Loop Brain Stimulation – Pushing the Boundaries towards Clinical Applications through Multiple Modalities"
6. Oral Presentation — Australasian Brain Stimulation Society (ABSS), Sydney, Australia (Nov 2023). "Using tDCS to improve attention in children with acquired brain injury: Correlates of response and home-based treatment"

### Oral and Poster Presentations

7. Oral Presentation – Neuromodec NYC Neuromodulation (Aug 2026): "Establishing Transcranial Endogenous Current Stimulation (tECS): A Pilot Study Modulating the EEG Aperiodic Exponent with Biomimetic Stimulation"
8. Poster Presentation – SOBP Conference, NYC, USA (Apr 2026). "The Time Course of Neural Target Engagement in a Randomized Trial of 10 Hz Tacs for Depression: Daily EEG Alpha Network Dynamics across Five Days"
9. Poster Presentation – SOBP Conference, Toronto, Canada (Apr 2025). "Daily changes in high-density EEG alpha power and functional connectivity across five consecutive days of 10Hz transcranial alternating current stimulation in depression: Results from a randomized controlled trial"
10. Poster Presentation — 5th International Brain Stimulation Conference, Lisbon, Portugal (Feb 2023). "tDCS at home in children with acquired brain injury (hrtDCS-Attention): A dose-controlled clinical trial protocol"

**Manuscripts**

1. **Stein A**, Floegel E, Zhang M, Subramaniyan A, Sharma S, Feldman Z, Evers M, Riddle J, Meltzer-Brody S, Frohlich F. (2026). "Establishing naturalistic brain stimulation targeting aperiodic EEG features: Transcranial Endogenous Current Stimulation (tECS)". *Brain Stimulation: Basic, Translational, and Clinical Research in Neuromodulation*.  
<https://doi.org/10.1016/j.brs.2026.103036>
  - Role: Conceptualization, developed novel stimulation technique, analyzed all data (high-density EEG power, connectivity analyses), supervised data collection (primary mentor for trial team), wrote original draft
2. Friehs, M.A., Schroeder, P.A., Barlow, K.M., **Stein, A.** (2026). "Not just childish games: An exploration of different game-based assessments and discussion of gamification in clinical pediatric populations". *J Cogn Enhanc*.  
<https://doi.org/10.1007/s41465-026-00345-9>
  - Role: Conceptualization, collected all data, analyzed all data (task performance, clinical outcomes), wrote 50% of original draft
3. **Stein A**, Riddle J, Caulfield KA, Dux PE, Friehs MA, Schroeder PA, Craven MP, Groom MJ, Iyer KK, Barlow KM (2025). "Safety, Feasibility, and Tolerability of Ten Days of At-Home, Remotely Supervised tDCS During Gamified Attention Training in Children with Acquired Brain Injury: An Open-Label, Dose-Controlled Pilot Trial." *Brain Sciences*. 2025 May 24;15(6):561. <https://doi.org/10.3390/brainsci15060561>
  - Role: Conceptualization, collected all data, analyzed all data (HD-EEG source localized connectivity, clinical outcomes), wrote original draft
4. **Stein A**, Caulfield KA, Singh M, Riddle J, Friehs MA, Craven MP, Groom MJ, Iyer KK, Barlow KM (2025). "The Effect of a Single Session of Transcranial Direct Current Stimulation on Attention in Pediatric Acquired Brain Injury: Characterizing Interindividual Structural and Functional Network Response Variability. *Pediatric Neurology*. 2025 Sep;170:133-45. <https://doi.org/10.1016/j.pediatrneurol.2025.06.005>
  - Role: Conceptualization, collected all data, analyzed all data (HD-EEG source localized connectivity, clinical outcomes), wrote original draft
5. **Stein A\***, Barlow KM\* (2025). "Orthostatic tachycardia in children with and without persisting post concussion symptoms following mild traumatic brain injury: A prospective controlled study". *Pediatric Neurology*. 2025 Nov 7. <https://doi.org/10.1016/j.pediatrneurol.2025.11.001>
  - Role: Analyzed all data (physiological measurements, clinical outcomes), wrote original draft
6. **Stein A**, Higgins N, Gajwani M, Gericke CA (2024). "The treatment gap for deep brain stimulation in Parkinson's disease: a comparative analysis of cost and utilisation in high-income countries". *Australian Health Review*. 2024 Sep 16;48(5):497-505. <https://doi.org/10.1071/AH24036>
  - Role: Collected all data (secondary data), analyzed data (cost, utilization, access data comparative analysis), wrote original draft
7. **Stein, A.**, Thorstensen, J.R., Ashley, D.P., Ho, J. M., Iyer, K. K., Barlow, K. M. (2024). "Attention please! Unravelling the link between brain network connectivity and cognitive attention following acquired brain injury: a systematic review of structural and functional measures." *Brain Connectivity*. <https://doi.org/10.1089/brain.2023.0067>
  - Role: Conceptualization, led data collection (systematic review), analyzed all data (systematic review and quality analysis), wrote original draft
8. **Stein, A.**, To, X.V., Nasrallah, F.A., Barlow, K.M. (2023). "Evidence of ongoing cerebral microstructural reorganization in children with persisting symptoms following mild traumatic brain injury: a NODDI DTI analysis." *J Neurotrauma*.  
<https://doi.org/10.1089/neu.2023.0196>
  - Role: Analyzed all data (diffusion tensor imaging, predictive models, clinical outcomes), wrote original draft
9. **Stein, A.**, Iyer, K.K., Barlow, K.M. (2022). "The effect of pediatric traumatic brain injury on simulated transcranial direct current stimulation electric field" [conference paper]. *Neuroergonomics and Neuromodulation Meeting Jul 2022: New York City, New York*. <https://shorturl.at/2fj1O>
  - Role: Analyzed all data (electric field modelling), wrote original draft
10. **Stein, A.**, Iyer, K.K., Khetani, A., Barlow, K.M. (2021) "Changes in Working Memory-Related Cortical Responses Following Pediatric Mild Traumatic Brain Injury: A Longitudinal Study". *J Concussion*, 5, p 1-15.  
<https://doi.org/10.1177/20597002211006541>
  - Role: Analyzed all data (task functional MRI, predictive models, clinical outcomes), wrote original draft

## Book chapters

11. **Stein, A.**, & Frohlich, F. (in press). "Making waves: Overview and methodological considerations for the use of tACS in clinical and cognitive neuroscience applications". In: E. G. Chrysikou (Ed.) Neuroscience Methods. New York: Springer Nature.
  - Role: wrote 80% of original draft (review paper)
12. **Stein, A.**, Iyer, K. K., & Barlow, K. M. (2022). "Cerebral activation of attention and working memory in traumatic brain injury." In: Diagnosis and Treatment of Traumatic Brain Injury. Academic Press. p 151-167.  
<https://doi.org/10.1016/B978-0-12-823347-4.00025-7>
  - Role: wrote original draft (review paper)

## Preprints

13. **Stein A\***, Schwippel TU\*, Pupillo FM, LaGarde HC, Zhang M, Rubinow DR, Frohlich F (2026). "Daily EEG reveals stage-specific alpha power and functional connectivity modulation across five days of tACS in major depressive disorder." medRxiv. 2026:2026-03. <https://doi.org/10.64898/2026.03.17.26348546>
  - Role: Analyzed data (high density EEG power and connectivity analysis, behavioral outcomes), wrote original draft
14. Schwippel T, Pupillo F, LaGarde H, **Stein A**, Zhang M, Rubinow D, Frohlich F (2026). "Suppression of Endogenous Alpha Power Predicts Clinical Response to 10 Hz tACS in Major Depressive Disorder: A Double-Blind Randomized Controlled Trial." medRxiv. 2026:2026-03. <https://doi.org/10.64898/2026.03.17.26348625>
  - Role: Review and editing

## Mentoring and Supervision

|                                                                                                                                                                                               |                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Carolina Center for Neurostimulation, UNC-CH</b>                                                                                                                                           | <b>Jul 2024 – present</b> |
| Lead mentor and supervisor for a clinical research team consisting of 9 undergraduate research assistants and 3 full-time study coordinators, across three brain stimulation clinical trials. |                           |
| <b>Co-advisor: PhD student, UQ Acquired Brain Injury in Children research group</b>                                                                                                           | <b>Aug 2024 - present</b> |
| Efficacy of at-home tDCS to improve attention in children with acquired brain injury                                                                                                          |                           |
| <b>Co-advisor: Honors student, UQ Acquired Brain Injury in Children research group</b>                                                                                                        | <b>Jan 2024 -Dec 2024</b> |
| Analysis of brain connectivity in 6 year-old children born pre-term. <u>Student outcome: First Class Honors</u>                                                                               |                           |

## Service, Outreach, and Volunteering

### Memberships

|                                                             |                       |
|-------------------------------------------------------------|-----------------------|
| • Associate Member, Society for Research in Psychopathology | <b>2025-present</b>   |
| • Member, American Australian Association                   | <b>2024 – present</b> |
| • Member, Graduate Women in Science (GWIS)                  | <b>2024 – present</b> |
| • Member, CuttingEEG                                        | <b>2023 – present</b> |
| • Member, Australasian Brain Stimulation Society (ABSS)     | <b>2022 – present</b> |
| • Fellow, Higher Education Academy (FHEA)                   | <b>2022 – present</b> |

### Professional Service

|                                                       |                              |
|-------------------------------------------------------|------------------------------|
| • UQ Rep, Biological Psychiatry Australia Committee   | <b>Jan. 2022 – Dec. 2022</b> |
| • Chair, UQ Faculty of Medicine HDR Student Committee | <b>Jan. 2021 – Dec. 2021</b> |
| • Co-Chair, UQ CCHR HDR Student Committee             | <b>Jan. 2021 – Dec. 2021</b> |

### Science Communication

|                                                                     |                             |
|---------------------------------------------------------------------|-----------------------------|
| • Music and the brain (ongoing events – UNC)                        | <b>Dec. 2024 – present</b>  |
| • Science Ambassador – Wonder of Science QLD (UQ)                   | <b>Mar. 2021 – 2024</b>     |
| • Science Blog – <a href="#">A Stimulating Endeavour</a> (Personal) | <b>Sep. 2020 – Sep 2024</b> |

### Community Outreach

|                                                                                                                                                                                  |                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| • <b>RSPCA</b> – Animal care assistant                                                                                                                                           | <b>Dec 2023 – Jul 2024</b>   |
| • <b>Wesley Mission Asher House</b> – Care Volunteer, Acquired brain injury (Reference: Marie Duffy, <a href="mailto:MDuffy@wmq.org.au">MDuffy@wmq.org.au</a> )                  | <b>Feb. 2019 – Nov. 2019</b> |
| • <b>Queensland Children's Hospital</b> – Bedside Play Volunteer (Reference: Nicky Pearson, <a href="mailto:nicky.pearson@childrens.org.au">nicky.pearson@childrens.org.au</a> ) | <b>Aug. 2017 – Feb. 2018</b> |

## Media engagement/Science Communication

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### **The Rhythmic Forum, Carolina Center for Neurostimulation**

**April 2026**

I led coordination of, and performed in, this [music and neuroscience mashup event](#) on behalf of the Carolina Center for Neurostimulation featuring several musical performances, a neuroscience/mental health panel, and live EEG. Collaboration with Carolina Performing Arts and UNC School of Medicine.

### **Neuroscience and music event, Carolina Performing Arts/UNC School of Medicine**

**Feb 2025**

I led and coordinated the participation of Carolina Center for Neurostimulation in [an event attracting an audience of over 200](#) featuring live EEG streaming during a violin performance by MacArthur Genius Fellow Johnny Gandelman.

### **Interviewee/media article subject, UNC Health/School of Medicine**

**Dec 2024**

Interviewed for a feature article and video interview on 'Postdoc Journeys' at UNC Health/School of Medicine.

### **TV Interviewee, 10 News Australia**

**Dec 2023**

Participated in a [TV news interview](#) about a publication using advanced DTI (NODDI) to investigate structural connectivity during recovery from pediatric mild TBI.

### **Media article subject, "Advanced imaging predicts the recovery of children with brain injury"**

**Dec 2023**

Published in [UQ Queensland Brain Institute Media Release](#) and [The Hippocratic Post](#)

### **TV interview, 10 News Australia**

**Oct 2023**

I participated in a [TV news interview](#) about my personalised home tDCS headband and home-based clinical trial using tDCS to improve attention following acquired brain injury in children. This interview was conducted to promote entries into the Bionics Gamechangers Australia Student Innovation Challenge.